

Ex. No 6: The Sleuth Kit (TSK): Command-Line Digital Forensics

Date:

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To use The Sleuth Kit (TSK), a collection of command-line tools, to analyze a disk image (4Dell Latitude CPi.E01), examine its file system, and recover digital evidence.

Prerequisites & Installation

- **1. Sleuth Kit Tools:** Download the Windows version of Sleuth Kit (sleuthkit-4.14.0-win32.zip) and extract it to a known location (e.g., C:\Users\sleuthkit-4.14.0-win32).
- **2. Evidence Files:** Download the sample disk image files (4Dell Latitude CPi.E01 and 4Dell Latitude CPi.E02) from the lab repository.
- **3. Case Folder:** Create a dedicated folder for your investigation, for example: C:\Forensics_Lab. Place the evidence files inside this folder.

Procedure

Step 1: Open Command Prompt & Navigate

Open the Command Prompt (cmd.exe) as Administrator.

Navigate to the bin directory inside your Sleuth Kit folder where all the executable tools (.exe) are located:

```
C:\Windows\System32> cd C:\Users\sleuthkit-4.14.0-win32\bin
```

```
Microsoft Windows [Version 10.0.26100.6899]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Windows\System32>cd C:\Users\sleuthkit-4.14.0-win32\bin
```

Step 2: List Partitions (mmls)

Use the mmls (list partitions) command to view the partition table of the disk image to identify the starting sector offset of the target partition:

```
C:\Users\sleuthkit-4.14.0-win32\bin> mmls.exe "C:\Forensics_Lab\4Dell Latitude CPi.E01"
```

Tip: The output shows the NTFS / exFAT (0x07) partition (Slot 002) starts at sector 63. This is our offset for the next commands.

```
Microsoft Windows [Version 10.0.26100.6899]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Windows\System32>cd C:\Users\sleuthkit-4.14.0-win32\bin
```

```
C:\Users\sleuthkit-4.14.0-win32\bin>mmls.exe "C:\Forensics_Lab\4Dell Latitude CPi.E01"
```

```
DOS Partition Table
```

```
Offset Sector: 0
```

```
Units are in 512-byte sectors
```

	Slot	Start	End	Length	Description
000:	Meta	0000000000	0000000000	0000000001	Primary Table (#0)
001:	-----	0000000000	0000000062	0000000063	Unallocated
002:	000:000	0000000063	0009510479	0009510417	NTFS / exFAT (0x07)
003:	-----	0009510480	0009514259	0000003780	Unallocated

Step 3: Analyze File System (fsstat)

Use the fsstat (file system statistics) command to extract detailed information about the partition using the -o 63 offset flag:

```
C:\Users\sleuthkit-4.14.0-win32\bin> fsstat.exe -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01"
```

```
C:\Users\sleuthkit-4.14.0-win32\bin>fsstat.exe -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)  Size: 48-72  Flags: Resident
$ATTRIBUTE_LIST (32)       Size: No Limit  Flags: Non-resident
$FILE_NAME (48)            Size: 68-578   Flags: Resident,Index
$OBJECT_ID (64)            Size: 0-256    Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)          Size: 2-256    Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)                Size: No Limit  Flags:
$INDEX_ROOT (144)          Size: No Limit  Flags: Resident
$INDEX_ALLOCATION (160)     Size: No Limit  Flags: Non-resident
$BITMAP (176)              Size: No Limit  Flags: Non-resident
$REPARSE_POINT (192)       Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)      Size: 8-8     Flags: Resident
$EA (224)                  Size: 0-65536   Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536   Flags: Non-resident

C:\Users\sleuthkit-4.14.0-win32\bin>
```

Step 4: List Files and Directories (fls)

Use the fls (list files) command to recursively list all files and directories in the partition and redirect to a text file for documentation:

```
C:\Users\sleuthkit-4.14.0-win32\bin> fls.exe -r -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01" >
C:\Forensics_Lab\file_list.txt
```

Tip: Inspect file_list.txt in C:\Forensics_Lab to examine the discovered file entries.

```
+ d/d 5609-144-1: PCHEALTH
++ d/d 5610-144-6: HELPCTR
+++ d/d 5611-144-1: BATCH
+++ d/d 5726-144-6: Binaries
++++ r/r 5731-128-4: brpinfo.dll
++++ r/r 5727-128-4: HCAAppRes.dll
++++ r/r 5728-128-4: HelpCtr.exe
++++ r/r 5729-128-4: HelpHost.exe
++++ r/r 5730-128-4: HelpSvc.exe
++++ r/r 5732-128-4: msconfig.exe
++++ r/r 5733-128-4: msinfo.dll
++++ r/r 5734-128-4: notiflag.exe
++++ r/r 5735-128-4: pchshl_wa.cab
++++ r/r 5736-128-4: pchshell.dll
++++ r/r 5737-128-4: pchsvc.dll
+++ d/d 6335-144-6: Config
+++ d/d 6563-144-1: Cache
+++++ r/r 6561-128-5: Professional_32_1033.dat.bak
+++++ d/d 6562-144-1: CheckPoint
+++++ r/r 7254-128-5: Cntstore.bin
+++++ r/r 6342-128-4: dataspec.xml
+++++ r/r 5612-128-5: SAFStore.xml
+++++ r/r 6621-128-5: sereg.xml
+++ d/d 6347-144-6: Database
+++++ r/r 6564-128-4: HCdata.edb
+++++ r/- * 0: newsgroups.cab
+++++ r/- * 0: pss.cab
+++++ r/- * 0: rcBuddy.cab
+++ d/d 6336-144-6: DataColl
+++++ r/r 11914-128-4: CollectedData_1.xml
+++++ r/r 11936-128-4: CollectedData_10.xml
+++++ r/r 11937-128-4: CollectedData_11.xml
+++++ r/r 11938-128-4: CollectedData_12.xml
+++++ r/r 11939-128-4: CollectedData_13.xml
+++++ r/r 11940-128-4: CollectedData_14.xml
+++++ r/r 11941-128-4: CollectedData_15.xml
+++++ r/r 11915-128-4: CollectedData_2.xml
+++++ r/r 11916-128-4: CollectedData_3.xml
+++++ r/r 11917-128-4: CollectedData_4.xml
+++++ r/r 11921-128-4: CollectedData_5.xml
+++++ r/r 11932-128-4: CollectedData_6.xml
+++++ r/r 11933-128-4: CollectedData_7.xml
+++++ r/r 11934-128-4: CollectedData_8.xml
+++++ r/r 11935-128-4: CollectedData_9.xml
+++++ r/r 11942-128-4: history_db.xml
+++ d/d 6353-144-1: HelpFiles
+++ d/d 6349-144-6: Indices
+++++ r/r 6612-128-4: merged.hhk
+++++ r/r 6613-128-4: scoped_2.hhk
```

Use the `istat` (inode statistics) command to get detailed metadata about a specific file, such as its MAC (Modified, Accessed, Changed) times. From file `list.txt`, target file `Mr. Evil.bmp` is identified with inode 9871:

```

C:\Users\sleuthkit-4.14.0-win32\bin> istat.exe -o 63 "C:\Forensics_Lab\63Dell Latitude CPI (2).E01" 9871
MFT Entry Header Values:
Entry: 9871          Sequence: 1
$LogFile Sequence Number: 36723336
Allocated File
Links: 2

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 318 ( )
Created:      2004-08-20 04:35:01.221337600 (India Standard Time)
File Modified: 2004-08-20 04:35:01.221337600 (India Standard Time)
MFT Modified:  2004-08-20 04:35:01.231352000 (India Standard Time)
Accessed:      2004-08-20 04:35:01.221337600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: Mr. Evil.bmp
Parent MFT Entry: 5663 Sequence: 1
Allocated Size: 0      Actual Size: 0
Created:      2004-08-20 04:35:01.221337600 (India Standard Time)
File Modified: 2004-08-20 04:35:01.221337600 (India Standard Time)
MFT Modified:  2004-08-20 04:35:01.221337600 (India Standard Time)
Accessed:      2004-08-20 04:35:01.221337600 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: MREVIL~1.BMP
Parent MFT Entry: 5663 Sequence: 1
Allocated Size: 0      Actual Size: 0
Created:      2004-08-20 04:35:01.221337600 (India Standard Time)
File Modified: 2004-08-20 04:35:01.221337600 (India Standard Time)
MFT Modified:  2004-08-20 04:35:01.221337600 (India Standard Time)
Accessed:      2004-08-20 04:35:01.221337600 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-2) Name: N/A Resident size: 90
Type: $FILE_NAME (48-3) Name: N/A Resident size: 90
Type: $DATA (128-4) Name: N/A Non-Resident size: 6976 init_size: 6976
5187099 5187100 5187101 5187102 5187103 5187104 5187105 5187106
5187107 5187108 5187109 5187110 5187111 5187112

```

Use the `icat` (inode cat) command to extract the contents of the file using its inode number 9871, saving it directly as a .bmp image:

[illegible]

Step 7: Create an Event Timeline (mactime)

Create a full timeline of all file activity on the system. First, generate a body file using fls:

```
C:\Users\sleuthkit-4.14.0-win32\bin> fls.exe -m / -r -o 63 "C:\Forensics_Lab\4Dell Latitude CPi.E01" > C:\Forensics_Lab\body.txt
```

Second, use the mactime.pl Perl script to sort the body file into a chronological timeline:

```
C:\Users\sleuthkit-4.14.0-win32\bin> mactime.pl -b C:\Forensics_Lab\body.txt > C:\Forensics_Lab\timeline.txt
```

```
C:\Users\sleuthkit-4.14.0-win32\bin>
[main 2025-10-27T08:07:52.049Z] updateURL https://api2.cursor.sh/updates/api/update/win32-x64/cursor/1.7.44/c3c1f9f91fe32f3f385cf6a85146cd829097bfd4f971d12db3cdfc35742429b4/stable
[main 2025-10-27T08:07:52.050Z] update#setState idle
[main 2025-10-27T08:08:22.064Z] update#setState checking for updates
[main 2025-10-27T08:08:26.419Z] update#setState downloading
```

Rubrics

Criteria	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result

By following these steps, we successfully used The Sleuth Kit (TSK) to analyze the 4Dell Latitude CPi.E01 disk image. We were able to inspect the detailed metadata of a target file (istat on inode 9871), successfully recover the file Mr. Evil.bmp (icat), and generate a full chronological timeline of all file system activity (mactime). All output files and recovered evidence are stored in C:\Forensics_Lab, ready for reporting.